26 **C** de Broglie wavelength $\lambda = h/mv$, so $\lambda \propto 1/m$ for same speed v.

Since mass of electron is much smaller than proton, $\lambda_e > \lambda_p$

For single slit diffraction pattern, $b\sin\theta = \lambda$, so $\sin\theta \propto \lambda$, $\rightarrow$ diffraction angle for electron is much larger than protons.

27 **B** Electrons: $dN_e/dt = I / e = 2.0\times10^{-6} / 1.6\times10^{-19}$